A-A-50732
15 December 1987

COMMERCIAL ITEM DESCRIPTION

TEST SET, INTERFACE BUS (PMEL)

The General Services Administration has authorized the use of this Commercial Item Description

Scope. This Commercial Item Description (CID), describes a Bus Interface Tester (PMEL). This CID is meant as a minimum requirement for the Bus Interface Tester (PMEL) in which only those manufacturers that meet or surpass the following requirements are qualified per this CID. This Tester will be used to test parallel/BCD, IEEE-488, RS-232-C and CIIL interface equipment.

Classification. Equipment covered by this specification may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Type III, Class 6, Style E and Color R in accordance with the latest MIL-T-28800. This CID makes provision for the bid sample testing in accordance with the latest MIL-T-28800.

Applicable Documents.

Specifications and Standards. The following specifications and standards form a part of this CID to the extend specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

(Copies of specifications, standards, and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

AMSC N/A FSC6625

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Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS

IEEE 587-1980; Surge Voltage in Low-Voltage AC Power Circuits, Guide for
IEEE 716-1982 c/Atlas Test Language - 18 Dec 81

(Application for copies of IEEE publications should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10018)

Order of precedence. In the event of conflict between the text of this CID and the references cited herein (except for the associated detail specifications, specification sheets for MS standards) the text of this CID shall take precedence. Nothing in this CID, however shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the Type III safety requirements of MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio Frequency Interference (RFI) Capacitors. Any capacitor used in RFI filters shall be rated for at least four (4) times the operating voltage.

c. Type III Ventilation Openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials, and processes. Except for marking, the Bus Interface Tester shall comply with the Type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with the latest MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

a. Restricted Materials. The equipment shall comply with the restricted material requirements of the latest MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

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3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the Type III design and construction requirements of the latest MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Type of construction. Construction of the Test Set, Interface Bus shall be of a test system design containing all components, firmware (software) to fully test the bus interface of all equipment and be expandable to other equipment with the availability of procurable firmware.

b. Signal input and output connectors. Shall be as stated in section 12.a of this CID.

c. Internal cooling. The equipment shall be capable of operating continuously for twenty-four (24) hours at 55°C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the Test Set, Interface bus when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts ac (±10%) at line frequencies of 50 Hz to 63 Hz.

a. Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

b. Power consumption. Maximum power consumption of all components combined shall not exceed 300 watts.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

d. Surge and transient voltage protection. The Test Set, Interface shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per Type III solid state tolerance bands of MIL-T-28800, the surge voltage

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protection circuit shall conform to IEEE 587-1980. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the Test Set, Interface Bus may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal Type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the Test Set, Interface Bus shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater than ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the Test Set, Interface Bus shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and weight. Dimension and weight is not a criteria for this CID.

6. Enclosure requirements. Unless otherwise specified herein, the equipment shall comply with the Style E enclosure requirements of MIL-T-28800.

a. Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

7. Marking and identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable work, phrase, or abbreviation hereof, indicating the purpose or use of the part.

(1) Panel marking. The front panel marking shall comply with the Type III parts, materials and processes panel marking requirements of MIL-T-28800; however, marking identification processes other than the Type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the Class 6 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in a non-operating environment of 10°C to +50°C.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 10°C to 40°C. Exception may be made on the system printer.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for Class 6 equipment as specified by MIL-T-28800, except the humidity range shall be 20 to 80% minimum.

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c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for Class 6 equipment as specified by MIL-T-28800.

d. Vibration requirements. The vibration requirements are waivered.

e. Mechanical shock requirements.

(1) Shock, pulse. The shock, pulse requirements are waivered.

(2) Bench handling. The bench handling requirements are waivered.

(3) Shock, high impact. The shock, high impact requirements are waivered.

f. Electromagnetic Interference (EMI). The equipment shall comply with standard commercial EMI practices.

g. Magnetic environment. The magnetic environment requirements are waivered.

9. Reliability. Reliability verification testing, shall be in accordance with the Type III, Class 6 environmental conditions of MIL-T-28800.

a. Mean Time Between Failures (MTBF). The equipment Mean Time Between Failures (MTBF) shall be 2,000 hours.

b. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40°C for a minimum of 48 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a Satisfactory Operation Test (SOT) shall be performed. A Satisfactory Operation Check (SOC) shall be performed conducted at selected times during the burn-in test.

(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycle for Class 6 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling. A Satisfactory Operation Check (SOC) shall be conducted during the last two upper and lower temperature limits. The first cycle high shall be 60°C.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

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a. Maintainability. The equipment shall meet the Type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the Type III interchangeability requirements specified by MIL-T-28800.

12. Performance characteristics. The Test Set, Interface Bus shall include and not be limited to all equipment and application testing listed herein.

a. Controller. The tester shall contain a computing instrument controller. The controller shall include the following:

(1) High resolution graphic CRT display, touch sensitive or keyboard operative.
(2) Program development system.
(3) Basic Interpreter.
(4) 2 MIPS processor.
(5) Macrostore high-speed floating point processor.
(6) 136K bytes read/write memory.
(7) ATLAS/Flex System package.
(8) Detachable programmer's keyboard.
(9) Expansion slots (minimum 2).
(10) 400K byte floppy disk.
(11) Real-time non-volatile calendar/clock.
(12) Parallel/BCD interface.
(13) IEEE-488 interface.
(14) RS-232-C interface.
(15) Ability to convert IEEE Standard 716 1982 (C/ATLAS) and T.O. CFE 48 (MATE ATLAS Extensions) to CIIL language. This will be in accordance with Proposed Standard No. 2806763 rev c.
(16) RS-232-C printer interface.
(17) IEEE-488 hard disk interface.
(18) 1M byte RAM expansion.
(19) All firmware (software) for fully test of the interface bus of all equipment.

b. CRT display. The CRT display shall be a high-contrast green phospher, high resolution display. The display format shall be selectable for 16 lines of 80 characters or 8 lines of 40 double-size characters.

c. Scanning method. The CRT shall utilize a non-interlaced raster scan.

d. Character memory. There shall be 1280 bytes of dedicated display memory.

e. Character set. The character set shall contain a minimum of 96 standard ASCII characters, math characters and graphic characters.

f. Enhancements. The CRT display shall contain reverse video, blinking, underlining, and highlighting character enhancement.

g. Cursor. The cursor shall be selectable for blinking, underline, block, or suppressed.

h. Touch sensitive. The CRT shall be of a touch sensitive operator interface or operator interface via keyboard functions keys.

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i. Keyboard. The programmer's keyboard shall be QWERTY style, detachable unit. The keyboard shall be designed as to where the keys shall not be damaged or inadvertently pressed when the keyboard is placed upside down. The keyboard shall generate the full ASCII character set plus all required function keys.

j. IEE-488 inteface handshake repertoire. The IEEE-488 shall contain the following functions.

(1)	Source Handshake	SH1
(2)	Acceptor Handshake	AH1
(3)	Basic Talker	T6
(4)	Basic Listener	L4
(5)	Controller	C1 thru C4, C6
(6)	Device Trigger	DT1
(7)	Device Clear	DC1
(8)	Service Request	SR1
(9)	Extended Talker	TE
(10)	Extended Listener	LE
(11)	Remote/Local	RL

k. IEE-488 remote message capability. The controller shall communicate with instruments by using the following:

(1)	Attention	ATN
(2)	Data Byte	DAB
(3)	Data Accepted	DAC
(4)	Data Valid	DAV
(5)	Device Clear	DCL
(6)	Interface Clear	IFC
(7)	My Listen Address	MLA
(8)	My Talk Address	MTA
(9)	Remote Enable	REN
(10)	Ready for Data	RFD
(11)	Unlisten	UNL
(12)	Untalk	UNT

l. RS-232-C interface. The controller shall come with RS-232-C interface with the following settable parameters for each RS-232-C port:

(1) Baud rate (75-19,200 baud)
(2) Data bits per character
(3) End of file character
(4) End of line character
(5) Parity
(6) Stop bits
(7) Stall input enable/disable (XON/XOFF)
(8) Stall output enable/disable (XON/XOFF)
(9) Time out

m. Parallel interface. The controller shall be able to bidirectional transfer monitor and control status bits, 16-bit words, or 512-word blocks via parallel interface. The following handshake or strobe protocols may also be selected:

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(1) Status Input/Output
(2) Full handshake input
(3) Full handshake output
(4) Pulse handshake input
(5) Pulse handshake output
(6) Strobe input
(7) Strobe output

n. Real time clock. The controller shall be supplied with a battery back-up read-time clock. The clock shall supply year, month, day, hour, minute and second with leap-year handling. The clock accuracy shall be a minimum of 3 sec/day.

o. Support. The controller shall contain all support equipment to test all equipment, but shall not be limited to the listing.

p. Documentation. The controller shall come with all development and testing documentation for support.

q. Hard disk drive. The Test Set, Interface Bus shall come with a 20M byte hard disk drive. The disk drive shall have a transfer rate of 160K byte/second with on line capacity of 20M bytes.

r. Printer. The bus interface tester printer shall be a dot-Matrix type or ink-jet type with a print speed of 100 characters per second. The printer shall operate bidirectional. The paper width shall be 4.5 inches to 10 inches wide. The interface shall be serial RS-232 with baud rates from 110 to 4,800 bits per second. The minimum buffer capacity shall be 2,000 characters with XON/XOFF protocol.

s. Manuals. The Test Set, Interface Bus shall come with two (2) sets of operating, instruction, programming and documentation manuals.

t. Cables. The Test Set, Interface Bus shall come with one (1) RS-232C interface cable 2M long, one (1) serial printer cable 2M long, and two (2) IEEE-488 interface cable 2M long.

13. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, and standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.)

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Known Acceptable Products. The following products are known to have met the requirements of this CID certification but submit to bid testing and certification as required.

Manufacturer	Model	NSN
Fluke	A1002	

Custodian
Air Force - 99
Navy - EC
DLA - ES

Review Activity
Army - CR

Preparing Activity:
Air Force - 82

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